Complete Blood Count (CBC) With Differential, Reflex to Peripheral Smear Review

Test Details

METHODOLOGY

Automated cell counter with mixed technologies

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Test Includes

Hematocrit; hemoglobin; mean corpuscular volume (MCV); mean corpuscular hemoglobin (MCH); mean corpuscular hemoglobin concentration (MCHC); red cell distribution width (RDW); percentage and absolute differential counts; platelet count (PLT); red cell count (RBC); white blood cell count (WBC)

Use

As a screening test to evaluate overall health; detect and/or identify a wide range of hematologic disorders; assist in managing medications/chemotherapeutic decisions

Custom Additional Information

Using advanced flow-cytometric technology, our sophisticated instruments will analyze the specimen based upon a complex set of flags and middleware rules. These rules help separate relatively normal results with nonspecific abnormalities from patients with hemolysis, hematological malignancies and/or the presence of microorganisms. Normal CBCs with differentials and CBCs with nonspecific abnormalities (i.e. nonspecific neutrophilia, anemia, thrombocytopenia) will be reported without smear review. The absence of additional report comments indicates that a smear review was not medically necessary. If specific morphologic abnormalities are detected, a peripheral smear will be prepared for review by a medical laboratory scientist. Depending on the abnormality, the medical laboratory scientist may perform a manual differential and/or assess for red cell morphology (abnormalities such as basophilic stippling, Howell-Jolly bodies, target cells, spherocytes and schistocytes), white cell abnormalities (neutrophil hypersegmentation, hypogranularity, toxic granulation, Dohle bodies or abnormal cells), or platelet abnormalities (i.e., hypogranular, large or giant platelets). If there are abnormal cells such as blasts, or lymphoma cells, microorganisms or evidence of hemolysis, the smear will be examined by a pathologist. The pathologist's comments will appear in a footnote to the CBC or in a separate report and will detail the hematologic disease. Life-threatening hematologic disorders (i.e., acute leukemia, chronic myeloproliferative disorders, microangiopathic hemolytic anemia, bacteremia) also will be called to the physician as a critical value.

The CBC with differential cascade will be performed as follows: 1) Run CBC through analyzer. 2) Apply flags and rules to generate smear. 3) Medical laboratory scientist reviews smear. 4) If criteria met, pathologist reviews smear.

Please note that many CBC abnormalities are nonspecific and not definitively related to a hematologic disorder. Please refer to the CBC Guide for the differential diagnosis of these CBCs.

A six-part differential reported in some lab locations includes IG% and IG absolute counts. IG (immature granulocytes) includes metamyelocytes and myelocytes. It does not include bands or blast cells. Promyelocytes and blasts are reported separately to denote the degree of left shift. An elevated percentage of IG has not been found to be clinically significant as a sole clinical predictor of disease. IGs are associated with infections, a variety of inflammatory disorders, cytokine therapy, neoplasia, hemolysis, tissue damage, seizures, metabolic abnormalities, myeloproliferative neoplasms and with the use of certain medications such as steroids.

Specimen Requirements

Specimen

Whole blood

Volume

Fill tube to capacity.

Minimum Volume

0.5 mL (500 µL for pediatric microtainer capillary tubes; fill tube to capacity) (**Note:** This volume does **not** allow for repeat testing.)

Container

Lavender-top (EDTA) tube

Collection Instructions

Invert tube 8 to 10 times immediately after tube is filled at the time of collection.

Stability Requirements

Temperature	Period
Room temperature	1 day
Refrigerated	3 days
Frozen	Unstable
Freeze/thaw cycles	Unstable

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Hemolysis; clotted specimen; specimen drawn in any anticoagulant other than EDTA; specimen diluted or contaminated with IV fluid; tube not filled with minimum volume; improper labeling; transfer tubes with whole blood; specimen received with plasma removed (plasma is used for other testing)